

Part A – Theory

1. What are Missing Values?

Missing values are data points that are not recorded or are unavailable in a dataset.

They are usually represented as:

- NaN (Not a Number)
- Null
- Blank cells

Causes of Missing Values

- Data entry errors
- Equipment malfunction
- Survey non-response
- Data corruption

Problems Caused by Missing Values

- Reduces data quality
- Produces inaccurate analysis
- Affects machine learning model performance

Example

Age BMI

25 24.5

30 NaN

45 28.3

In the second row, BMI is missing.

2. Types of Missing Data

MCAR (Missing Completely At Random)

Missing values occur completely randomly and are unrelated to any variable.

Example:

A patient's cholesterol value is missing due to a machine failure.

Characteristics

- No pattern exists
 - Least problematic type
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MAR (Missing At Random)

Missing values depend on another observed variable.

Example:

Older patients are less likely to report glucose levels.

Characteristics

- Missingness depends on other variables
 - Can be handled using advanced imputation methods
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MNAR (Missing Not At Random)

Missing values depend on the missing value itself.

Example:

Patients with very high cholesterol choose not to disclose their cholesterol levels.

Characteristics

- Most difficult type
 - May introduce bias
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3. Missing Value Analysis

Missing Value Analysis is the process of identifying and measuring missing data in a dataset.

Formula

$$\text{Missing Percentage} = \frac{\text{Number of Missing Values}}{\text{Total Number of Records}} \times 100$$

Purpose

- Understand data quality
 - Select suitable imputation techniques
 - Improve machine learning performance
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4. Mean Imputation

Mean Imputation replaces missing numerical values with the average value of the column.

Formula

$$\text{Mean} = \frac{\sum X}{N}$$

Example

Original Data:

20, 25, NaN, 35, 40

Mean:

$$\frac{20 + 25 + 35 + 40}{4} = 30$$

After Imputation:

20, 25, 30, 35, 40

Advantages

- Easy to implement
- Fast

Disadvantages

- Sensitive to outliers
 - Reduces variance
-

5. Median Imputation

Median Imputation replaces missing values with the middle value of the dataset.

Example

Original Data:

20, 25, NaN, 35, 40

Median:

30

After Imputation:

20, 25, 30, 35, 40

Advantages

- Not affected by outliers
- Suitable for skewed distributions

Disadvantages

- May distort relationships between variables
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6. Most Frequent Imputation

Most Frequent Imputation replaces missing categorical values with the most commonly occurring category.

Example

Original Data:

Male, Female, Male, NaN, Male

Most Frequent Value:

Male

After Imputation:

Male, Female, Male, Male, Male

Applications

- Gender
 - Region
 - Category labels
-

7. Missing Indicator

A Missing Indicator creates a new binary feature indicating whether a value was originally missing.

Example

BMI BMI_Missing

25 0

NaN 1

30 0

Benefits

- Preserves information about missingness
 - Useful in machine learning
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8. Random Sample Imputation

Random Sample Imputation replaces missing values using randomly selected values from existing observations.

Steps

1. Select non-missing values
2. Randomly sample values
3. Replace missing entries

Advantages

- Preserves data distribution
- Better than mean imputation in many cases

Disadvantages

- Results may vary each run
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9. KNN Imputation

KNN (K-Nearest Neighbors) Imputation fills missing values using similar records.

Working

1. Find nearest neighbors
2. Calculate average of neighboring values
3. Replace missing value

Advantages

- Produces realistic values
- Maintains relationships between variables

Disadvantages

- Computationally expensive
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10. MICE (Multiple Imputation by Chained Equations)

MICE is an advanced imputation technique that predicts missing values using relationships among variables.

Working

1. Initialize missing values
2. Build prediction models
3. Estimate missing values
4. Repeat multiple times

Advantages

- Highly accurate
- Preserves variable relationships

Disadvantages

- Time consuming
 - Computationally expensive
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11. What are Outliers?

Outliers are observations that differ significantly from the majority of the data.

Example

20, 22, 25, 24, 21, 120

Here:

120

is an outlier.

Effects of Outliers

- Distort statistical measures
 - Affect machine learning models
 - Increase variance
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12. Z-Score Method

Z-Score measures how many standard deviations a value is from the mean.

Formula

$$Z = \frac{X - \mu}{\sigma}$$

Where:

- X = value
- μ = mean
- σ = standard deviation

Rule

$$|Z| > 3$$

⇒ Outlier

Advantages

- Easy to calculate
 - Useful for normally distributed data
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13. IQR (Interquartile Range) Method

IQR measures the spread of the middle 50% of data.

Formula

$$IQR = Q3 - Q1$$

Lower Bound:

$$Q1 - 1.5(IQR)$$

Upper Bound:

$$Q3 + 1.5(IQR)$$

Values outside these limits are considered outliers.

Advantages

- Works well with skewed data
 - Not affected by extreme values
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14. Percentile Method

The Percentile Method identifies outliers using percentile thresholds.

Common thresholds:

- 1st Percentile
- 99th Percentile

Values below the 1st percentile or above the 99th percentile are treated as outliers.

Advantages

- Simple
 - Flexible
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15. Winsorization

Winsorization is an outlier treatment technique that replaces extreme values with boundary percentile values.

Example

Before:

20, 22, 25, 30, 75

After Winsorization:

20, 22, 25, 30, 45

Benefits

- Reduces effect of outliers
- Maintains dataset size
- Improves model performance

Applications

- Finance
 - Healthcare
 - Machine Learning
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Conclusion

Missing values and outliers are common data quality issues. Proper handling of missing values using techniques such as Mean, Median, KNN, and MICE improves dataset completeness. Outlier detection methods like Z-Score, IQR, and Percentile help identify unusual observations, while Winsorization effectively treats extreme values. These

preprocessing steps improve data quality and prepare datasets for reliable statistical analysis and machine learning models.